MA612: Fundamentals of Analytics / Statistical methods

Course Placement: It's a core course for master students of Agri Analytics program.

Credit Structure: 3-0-2-4

Three 50 minutes' lectures every week.

Two hours Lab every week

Course Details:

This course offers a comprehensive introduction to the fundamental principles of statistical analysis and data analytics, carefully designed for direct application within agricultural sciences. Students will develop skills in the systematic collection, management, analysis, and interpretation of various types of agricultural data, utilizing modern statistical software. The curriculum covers key topics such as **descriptive statistics, probability distributions, statistical inference, experimental design principles, regression analysis and time series analysis.** Emphasis will be placed on integrating real-world agricultural case studies and practical exercises throughout the course. The main goal is to enable students to shift from traditional, often guesswork-based farming methods toward evidence-based, data-driven, and predictive agriculture. Students are expected to be familiar with the Python programing language basics for agricultural data analysis.

Textbooks:

1. Statistics for Business & Economics - Anderson et al., Cengage Learning (2018)
2. Think Stats - Allen B Downey (2011)
3. Statistics Using Python - Oswald Campesato (2023)
4. A Text Book of Agricultural Statistics - R. Rangaswamy (2022)

Course Outcomes: After successful completion of this course, students will be able to

- Collect, organize, and summarize agricultural data effectively.
- Apply descriptive and inferential statistical methods to various agricultural datasets.
- Design and analyze agricultural experiments using appropriate statistical techniques.
- Utilize Python for data analysis and visualization.
- Acquire the skill to analyze data in a systematic and logical manner to answer important research questions arising in the field of agriculture
- Interpret statistical results to inform data-driven decisions in agriculture.
- Communicate statistical findings clearly and effectively
- Recognize the ethical considerations and limitations of statistical analysis in agri-analytics.

Assessment method: Quizzes/Assignments, project presentation, Lab work and semester examinations.

Grading Policy:

30% Quizzes/Assignments

30% Project/Tests/Viva

40% Final exam

This grading policy is subject to change and the final grading policy will be updated once the class starts.

Tentative Lecture Schedule

Sl.No.	Description	No. of Lectures
1.	**Introduction to Data and Statistics in Agriculture**	5
	Basics of Statistics: Population vs. Sample, Parameters vs. Statistics, Data types, Types of agricultural data: field data, remote sensing data, weather data, market data, soil data, livestock data. Data Collection and Organization: Sources of agricultural data (sensor networks, drones, satellite imagery, field surveys, government databases). Data Visualization and Exploratory Data Analysis: Data visualization and summarization with the help of tables and the important descriptive measures.	
2.	**Probability Theory and Distributions**	5
	Concepts of probability (both objective and subjective). Statistical interpretation of probability, Bayes' theorem and its applications in solving real-life decision problems, Standard probability distributions: Standard probability models (Binomial, Poisson, Uniform, Exponential, Gaussian), Probability distributions of transforms of random variables, Data generation from probability distributions using Monte Carlo method.	
3.	**Sampling Techniques & Distributions**	7
	Basic sampling concepts (population, sample, sampling frame, sampling unit) and probability sampling methods (simple random sampling, stratified sampling, systematic sampling, cluster sampling). Concept of sampling distribution. Central Limit Theorem and its importance. Sampling distribution of the mean. Illustration of sampling distribution using Monte Carlo simulation	
4.	**Statistical inference**	10
	Statistical estimation: Point and interval estimation, Methods of estimation (method of moments & maximum likelihood method) **Hypothesis testing:** Null and alternative hypotheses, Significance testing using p-value, parametric and non-parametric tests **Analysis of Variance:** One-Way ANOVA: Comparing means of three or more independent groups (e.g., different crop varieties). Two-Way ANOVA: Examining the effect of two independent categorical variables and their interaction (e.g., fertilizer type and irrigation level on yield). Post-hoc tests (e.g., Tukey HSD) for multiple comparisons.	
5.	**Correlation & Regression Analysis:**	6
	Concept of correlation: Strength and direction of relationship, Pearson's correlation coefficient (r), Spearman's rank correlation (for non-parametric data), Scatter plots for visualizing correlation. Introduction to regression: Predicting one variable from another, simple and multiple linear regression, Least Squares method for estimating regression coefficients, Coefficient of determination, Polynomial regression, Logistic regression	
6.	**Time Series Analysis**	5
	Components of time series data (trend, seasonality, cycle, irregularity), Basic forecasting methods (e.g., moving averages), Applications in agricultural market prediction, weather pattern analysis.	
7.	**Design of Experiments:**	4
	Principles of Experimental Design: Randomization, Replication, and Local Control, Importance of good experimental design in agricultural research. Common Experimental Designs in Agriculture: Completely Randomized Design (CRD), Randomized Block Design (RBD), Latin Square Design (LSD), Factorial Experiments, Split-plot and Strip-plot designs, Analysis of data from these designs using ANOVA.	
8.	**Bootstrap estimation of Bias and Standard Error** (if time permits)	3